

Day 50



Javascript
THE LANGUAGE OF THE WEB

Modules in JavaScript:

What Are Modules?

Modules are simply separate JavaScript files that you can import and export. They help you split your code into smaller, manageable parts.

Example:

- math.js → contains math functions
- main.js → uses those functions

This makes your project cleaner and easier to maintain.

Why Do We Need Modules?

- Avoids writing everything in one big file
- Helps reuse code
- Makes debugging easier
- Prevents name conflicts
- Makes code organized and professional

Types of Modules

1. ES6 Modules (import/export) — modern and recommended
2. CommonJS (require/module.exports) — used in Node.js (older)

We will focus on ES6 Modules.

How to Export Something

You can export:

- variables
- functions
- classes

Example (math.js)

```
export function add(a, b) {  
  return a + b;  
}  
export const PI = 3.14;
```

How to Import Something

Example (main.js):

```
import { add, PI } from "./math.js";  
console.log(add(2, 3));  
console.log(PI);
```

Important Rules for Modules

- Modules run in strict mode by default
- Imports must be at the top of the file
- File paths must include .js extension in browsers
- Each module has its own scope